

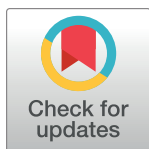
RESEARCH ARTICLE

Rhizosphere soil properties of waxy sorghum under different row ratio configurations in waxy sorghum-soybean intercropping systems

Mingbo Shao^{1*}, Can Wang, Lingbo Zhou, Fangli Peng, Guobing Zhang, Jie Gao, Siyu Chen, Qiang Zhao

Institute of Upland Food Crops, Guizhou Academy of Agricultural Sciences, Guiyang, Guizhou, China

* 563189433@qq.com



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Abstract

To overcome the continuous planting obstacle and promote the sustainable production of waxy sorghum, a two-years field experiment was performed to determine the responses of waxy sorghum rhizosphere soil properties to different row ratio configurations in waxy sorghum-soybean intercropping systems. The treatments included five row ratio configurations, which were two rows of waxy sorghum intercropped with one row of soybean (2W1S), two rows of waxy sorghum intercropped with two rows of soybean (2W2S), three rows of waxy sorghum intercropped with one row of soybean (3W1S), three rows of waxy sorghum intercropped with two rows of soybean (3W2S), and three rows of waxy sorghum intercropped with three rows of soybean (3W3S), and sole cropping waxy sorghum (SW) was used as control. The nutrients, enzyme activities, and microbes of waxy sorghum rhizosphere soil were investigated at the jointing, anthesis, and maturity stages. Results showed that rhizosphere soil properties of waxy sorghum were significantly affected by row ratio configurations of waxy sorghum intercropped soybean. Among all treatments, the performances of rhizosphere soil nutrients contents, enzymes activities, and microbes contents were 2W1S > 3W1S > 3W2S > 3W3S > 2W2S > SW. Compared to SW treatment, the 2W1S treatment increased the organic matter, total N, total P, total K, gram-negative bacteria phospholipid fatty acids (PLFAs), and gram-positive bacteria PLFAs contents and catalase, polyphenol oxidase, and urease activities by 20.86%–25.67%, 34.33%–70.05%, 23.98%–33.83%, 44.12%–81.86%, 74.87%–194.32%, 81.59–136.59%, 91.44%–114.07%, 85.35%–146.91%, and 36.32%–63.94%, respectively. Likewise, the available N, available P, available K, total PLFAs, fungus PLFAs, actinomycetes PLFAs, and bacteria PLFAs contents under the 2W1S treatment were 1.53–2.41, 1.32–1.89, 1.82–2.05, 1.96–2.91, 3.59–4.44, 9.11–12.56, and 1.81–2.71 times than those of SW treatment, respectively. Further, the determining factors of soil microbes were total K, catalase, and polyphenol oxidase for total microbes, bacteria, and gram-negative bacteria, total P and available K for fungus, available N, available K, and polyphenol oxidase for actinomycetes, and total K and polyphenol oxidase for gram-positive bacteria. In conclusion, the 2W1S treatment was the

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optimal row ratio configuration of waxy sorghum intercropped with soybean, which can improve the rhizosphere soil quality and promote the sustainable production of waxy sorghum.

Introduction

Sorghum [*Sorghum bicolor* (L.) Moench] is the fifth largest cereal crop second only to rice, wheat, corn, and barley, which is cultivated in many countries around the world [1]. It is an important miscellaneous grain crop in China and can be used for the human food, animal feed, and industrial raw materials [2, 3]. Waxy sorghum is the main raw material for brewing Moutai-flavor liquor and plays an important role in boosting rural revitalization of Guizhou province, China [4, 5]. According to the statistics, the annual demand of liquor factories for waxy sorghum planting area is about two hundred and fifty thousand hectares in Guizhou. However, due to the limited per capita arable land area, continuous planting of waxy sorghum has led to the deterioration of soil nutrients, unbalance of soil microbial community structure, intensification of diseases and pests, and reduction of yield and quality [6, 7]. Therefore, it is urgent to find a suitable planting pattern to overcome the continuous planting obstacle of waxy sorghum.

Intercropping refers to the planting method of two or more crop species simultaneously planted in the same field during the same growing season, which can improve the land productivity when compared with sole planting systems [8–10]. At present, explanations for the benefits of intercropping systems are mainly reflected in the following four aspects. Firstly, it can promote the soil improvement, especially in terms of the soil microbes. As an example, the tobacco intercropped with soybean significantly increased the amounts of bacteria and actinomycetes, while reducing the fungus amount in rhizosphere soil [11]. Secondly, it can use the complementary effect of different niche resources to promote the rooting behavior of crops. For instance, intercropping has been found to enhance root functional traits in 20 cm depth of soil [12]. Thirdly, intercropping can improve crops ventilation and light conditions through the allocation of tall and short crops, so as to increase the light utilization efficiency of crops. A typical example is oat intercropped with legumes enhanced chlorophyll content and net photosynthetic rate of oat and altered chlorophyll composition which contributed to the slower process of oat aging [13]. Fourthly, intercropping can use biodiversity to control pests and diseases. Ma et al. [14] presented an example for this aspect that faba bean intercropped with wheat significantly decreased the chocolate spot disease index and area under disease progress curve of faba bean.

Intercropping of cereals and legumes is a sustainable planting pattern with ecological and intensive characteristics, which can reduce the dependence of agricultural production on chemical nitrogen fertilizer and pesticides through nitrogen fixation of legumes [15–17]. Legumes provide an important strategy to alleviate the constraints caused by soil nitrogen limitations and improve crop productivity, which can fix atmospheric nitrogen to reduce soil erosion and improve the land use efficiency [18–20]. Such horticultural cultivation can also help in the reduction of drought and salt stresses resulting from bad land use efficiency [21]. Sorghum intercropped with soybean often obtains more compound yield and economic benefit than those of their sole crop because of its more efficient use of resources [22, 23]. Sorghum is a typical cereal and a C₄ crop with high photosynthetic capacity, because it can use very low CO₂ content in the intercellular space of leaves for photosynthesis [24]. Soybean is a legume

and C₃ crop which can enhance the efficient use of nitrogen by using its own nitrogen-fixing bacteria to fix nitrogen sources [25]. These bacteria can help in the improvement of plant growth and development, as well as the bioremediation of different types of pollutants [26]. Thus, sorghum-soybean intercropping may be an option for improving crop yield and quality.

Row ratio configuration is a very important management measure in the intercropping systems, which can change the field microclimate and regulate interspecies competition to affect yield formation of intercrops [27, 28]. Thus, appropriate row ratio configuration is essential to alleviate the competition among intercrops and improve the land use efficiency. Yin et al. [29] have shown that the ecological and economic benefits were significantly affected by waxy sorghum intercropped with soybean. Nevertheless, to our knowledge, the changes of waxy sorghum rhizosphere soil properties in the waxy sorghum-soybean intercropping systems have not been reported, especially regarding row ratio configuration. Therefore, a two-years field experiment was performed to determine the responses of waxy sorghum rhizosphere soil properties to different row ratio configurations in waxy sorghum-soybean intercropping systems.

Materials and methods

Experimental sites and experimental design

In 2019 and 2020, the field experiments were performed at the Guiyang test base of Guizhou Academy of Agricultural Sciences. Waxy sorghum cultivar “Qiangao-7” and soybean cultivar “Qiandou-7” were used in this study. The treatments included five row ratio configurations of waxy sorghum intercropped soybean, which were two rows of waxy sorghum intercropped with one row of soybean (2W1S), two rows of waxy sorghum intercropped with two rows of soybean (2W2S), three rows of waxy sorghum intercropped with one row of soybean (3W1S), three rows of waxy sorghum intercropped with two rows of soybean (3W2S), and three rows of waxy sorghum intercropped with three rows of soybean (3W3S), and sole cropping waxy sorghum (SW) was used as control. All treatments were arranged in a randomized complete block arrangement with three replicates. Detailed descriptions of experimental site, experimental design, and various field management practices were presented in our previous studies [30, 31].

Rhizosphere soil sampling

At the jointing, anthesis, and maturity stages, three planting holes of waxy sorghum were randomly selected in the middle strip of each plot, and roots were dug up using a small hoe. The loose surface soil on the roots was shaken off gently and rhizosphere soil which tightly adhered to roots was collected carefully with a brush. Rhizosphere soils from waxy sorghum plants in each plot were mixed to make a composite sample and sieved through a 2 mm mesh. Then, each composite rhizosphere soil sample was divided into three parts evenly, and a portion of which was air-dried for two weeks and stored at room temperature prior to nutrient analysis. Other two parts of these rhizosphere soil samples were immediately frozen in liquid nitrogen and then stored at 4°C and −80°C in a refrigerator for analyses of enzyme activity and microbe, respectively.

Measurement of soil nutrients

The rhizosphere soil nutrients were measured as described by Yang et al. [32]. The organic matter was assayed using the potassium dichromate dilution thermal method. Total N was determined using the Kjeldahl distillation method. Total P was measured using the

molybdenum-antimony-D-isoascorbic acid colorimetry after wet digestion with $\text{HClO}_4\text{-H}_2\text{SO}_4$. Total K was determined using the flame photometry after molten with NaOH. Available N was assayed using the alkaline hydrolysis diffusion method. Available P was extracted with $0.5 \text{ mol L}^{-1} \text{ NaHCO}_3$ solution and then measured using the molybdenum-antimony-D-isoascorbic acid colorimetry. Available K was extracted with $1 \text{ mol L}^{-1} \text{ NH}_4\text{OAc}$ neutral extraction and then determined by the flame photometry.

Measurement of soil enzyme activities

The catalase and polyphenol oxidase activities were assayed according to the method of Zhou et al. [33]. The urease activity was assayed according to the method of Wang et al. [34]. Catalase activity was assayed using the potassium permanganate titration method and one unit of enzyme activity (U) was defined as a consumption of KMnO_4 solution in 1 mL after 30 min. Polyphenol oxidase activity was measured using the pyrogallol colorimetry method and one unit of enzyme activity (U) was defined as a formation of purple pyrogallol in 1 mg after 2 h. Urease activity was determined using the phenol-sodium hypochlorite colorimetry method and one unit of enzyme activity (U) was defined as a formation of $\text{NH}_4\text{-N}$ in 1 mg after 24 h.

Determination of soil microbes

Rhizosphere soil microbes were determined through the phospholipid fatty acids (PLFAs) analysis according to the modification method of Luo et al. [35]. Briefly, 4 g of rhizosphere soil sample was extracted with 15.2 mL of extracting solution, which contained 3.2 mL of citric acid buffer, 4 mL of chloroform and 8 mL of methanol. The extracts were centrifuged at 4000 r min^{-1} for 10 min and the supernatants were evaporated under N_2 . Phospholipids were separated on silicic acid columns by sequentially eluting with organic solvents of increasing polarity, and then saponified and methylated to form fatty-acid methyl esters (FAMES). FAMES were dissolved with hexyl hydride and then quantified with 19:0 methyl ester internal standard using a gas chromatograph. Identification of FAMES was conducted using Sherlock Microbial Identification System 4.5. Individual PLFAs were used to indicate broad groups of the microbe community as follow: 12:0, 13:0, a13:0, i13:0, 14:0, a14:0, i14:0, a15:0, i15:0, i15:1G, 16:0, a16:0, i16:0, i16:1H, 16:02OH, 16:1 ω 5c, 16:1 ω 11c, 17:0, a17:0, i17:0, cy17:0, 17:1 ω 8c, 18:0, 11Me18:1 ω 7c, a19:0, i19:0, cy19:0 ω 8c, and 20:0 for bacteria [32, 36]; a16:0, 16:02OH, 16:1 ω 5c, 17:0, cy17:0, 17:1 ω 8c, 18:0, and cy19:0 ω 8c for gram-negative bacteria; 14:0, a14:0, i14:0, a13:0, a15:0, i15:0, i15:1G, i16:0, a17:0, and i17:0 for gram-positive bacteria; 18:1 ω 9c and 20:1 ω 9c for fungus [35, 37]; 10Me17:0 and 10Me18:0 for actinomycetes [38].

Data analysis

One-way analysis of variance (ANOVA) was performed using DPS v7.05 software, and means were tested by least significant difference at the $P < 0.05$ level ($\text{LSD}_{0.05}$). Figures were drawn by SigmaPlot 12.5 software. Pearson correlation coefficient and multiple stepwise regression equation were used to reflect the relationships between soil microbes and soil nutrients as well as soil enzyme activities.

Results

Rhizosphere soil nutrients of waxy sorghum

From jointing stage to maturity stage, the organic matter (Fig 1A and 1B) and total P contents (Fig 1E and 1F) increased and the decreased, total N content (Fig 1C and 1D) decreased gradually, while total K content (Fig 1G and 1H) decreased and then increased. Among all

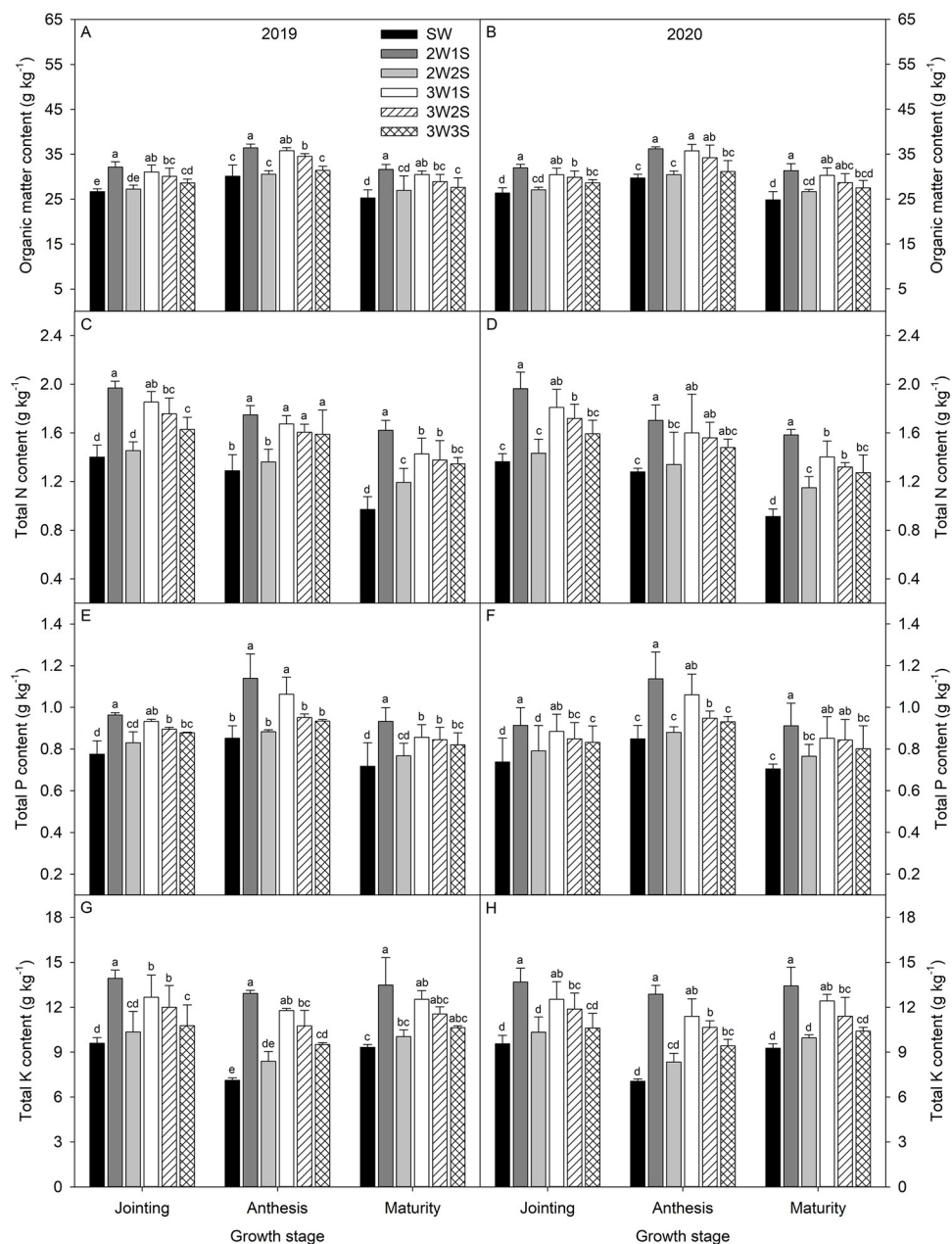


Fig 1. Rhizosphere soil organic matter (A, B), total N (C, D), total P (E, F), and total K (G, H) contents of waxy sorghum under different row ratio configurations in waxy sorghum-soybean intercropping systems. Data are expressed as mean $\pm$ SE ($n = 3$). Different letters indicate significant differences among treatments at the $P < 0.05$ level. SW represent sole cropping waxy sorghum. 2W1S, 2W2S, 3W1S, 3W2S, and 3W3S represent two rows of waxy sorghum intercropped with one row of soybean, two rows of waxy sorghum intercropped with two rows of soybean, three rows of waxy sorghum intercropped with one row of soybean, three rows of waxy sorghum intercropped with two rows of soybean, and three rows of waxy sorghum intercropped with three rows of soybean, respectively.

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treatments, the performances of organic matter, total N, total P, and total K contents were $2W1S > 3W1S > 3W2S > 3W3S > 2W2S > SW$ in each growth stage. Compared to SW treatment, the 2W1S treatment increased the organic matter, total N, total P, and total K contents by 20.86%, 42.25%, 23.98%, and 44.12% at the jointing stage, 21.42%, 34.33%, 33.83%, and

81.86% at the anthesis stage, and 25.67%, 70.05%, 29.58%, and 44.68% at the maturity, respectively.

The dynamic variation trend with growth stage for available N content (Fig 2A and 2B) was increased gradually, available P content (Fig 2C and 2D) was increased and the decreased, and

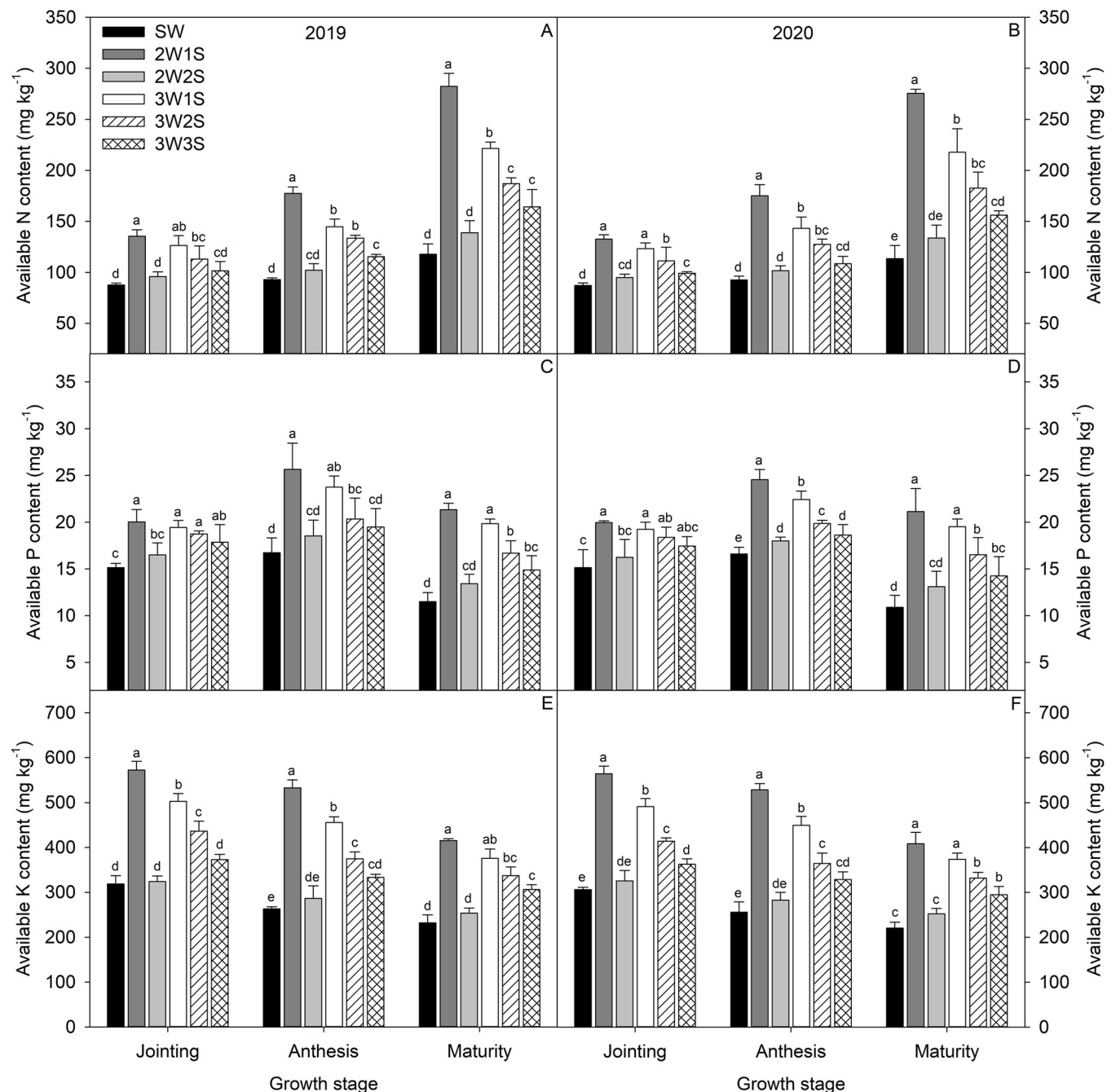


Fig 2. Rhizosphere soil available N (A, B), available P (C, D), and available K (E, F) contents of waxy sorghum under different row ratio configurations in waxy sorghum-soybean intercropping systems. Data are expressed as mean $\pm$ SE ($n = 3$). Different letters indicate significant differences among treatments at the $P < 0.05$ level. SW represent sole cropping waxy sorghum. 2W1S, 2W2S, 3W1S, 3W2S, and 3W3S represent two rows of waxy sorghum intercropped with one row of soybean, two rows of waxy sorghum intercropped with two rows of soybean, three rows of waxy sorghum intercropped with one row of soybean, three rows of waxy sorghum intercropped with two rows of soybean, and three rows of waxy sorghum intercropped with three rows of soybean, respectively.

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available K content (Fig 2E and 2F) was decreased gradually from jointing stage to maturity stage. Similarly, the available N, available P, and available K contents under all row ratio configurations of waxy sorghum intercropped soybean were higher than those of sole cropping waxy sorghum, and the 2W1S treatment showed the best performance. The available N, available P, and available K contents under the 2W1S treatment were 1.53, 1.32, and 1.82 times at the jointing stage, 1.90, 1.51, and 2.05 times at the anthesis stage, and 2.41, 1.89, and 1.82 times at the maturity stage than those of SW treatment, respectively.

Rhizosphere soil enzymes activities of waxy sorghum

From jointing stage to maturity stage, the dynamic variation trend for catalase activity (Fig 3A and 3B) was increased gradually, and polyphenol oxidase (Fig 3C and 3D) and urease activities (Fig 3E and 3F) were decreased and then increased. Among all treatments, the performances of catalase, polyphenol oxidase, and urease activities were $2W1S > 3W1S > 3W2S > 3W3S > 2W2S > SW$ in each growth stage. Compared to SW treatment, the 2W1S treatment increased the catalase, polyphenol oxidase, and urease activities by 114.07%, 128.65%, and 36.32% at the jointing stage, 95.77%, 85.35%, and 63.94% at the anthesis stage, and 91.44%, 146.91%, and 51.58% at the maturity, respectively.

Rhizosphere soil microbes of waxy sorghum

The dynamic variation trend with growth stage for total PLFAs (Fig 4A and 4B), actinomycetes PLFAs (Fig 4E and 4F), and bacteria PLFAs contents (Fig 4G and 4H) were decreased and then increased, while fungus PLFAs content (Fig 4C and 4D) was decreased gradually. The total PLFAs, fungus PLFAs, actinomycetes PLFAs, and bacteria PLFAs contents under all row ratio configurations of waxy sorghum intercropped soybean were higher than those of sole cropping waxy sorghum, and the 2W1S treatment showed the best performance. The total PLFAs, fungus PLFAs, actinomycetes PLFAs, and bacteria PLFAs contents under the 2W1S treatment were 1.96, 3.90, 9.11, and 1.81 times at the jointing stage, 2.91, 4.44, 12.56, and 2.71 times at the anthesis stage, and 2.55, 3.59, 9.91, and 2.27 times at the maturity stage than those of SW treatment, respectively.

Similarly, the gram-negative bacteria PLFAs and gram-positive bacteria PLFAs contents (Table 1) decreased and then increased with the growth process of waxy sorghum. The performances of gram-negative bacteria PLFAs and gram-positive bacteria PLFAs contents among all treatments were $2W1S > 3W1S > 3W2S > 3W3S > 2W2S > SW$ in each growth stage. Compared to SW treatment, the 2W1S treatment increased the gram-negative bacteria PLFAs and gram-positive bacteria PLFAs contents by 74.87% and 81.59% at the jointing stage, 194.32% and 136.59% at the anthesis stage, and 190.95% and 106.94% at the maturity stage, respectively.

Relationships between soil microbes and soil nutrients as well as soil enzyme activities

As shown in correlation analysis (Table 2), soil microbes were significantly correlated with soil nutrients and soil enzyme activities except for organic matter and total P contents with total PLFAs, bacteria PLFAs, gram-negative bacteria PLFAs, and gram-positive bacteria PLFAs contents, and available N content with fungus PLFAs and gram-negative bacteria PLFAs contents. Further, multiple stepwise regression analysis (Table 3) showed that the determining factors of soil microbes were total K, catalase, and polyphenol oxidase for total microbes, bacteria, and gram-negative bacteria, total P and available K for fungus, available N, available K, and polyphenol oxidase for actinomycetes, and total K and polyphenol oxidase for gram-positive bacteria.

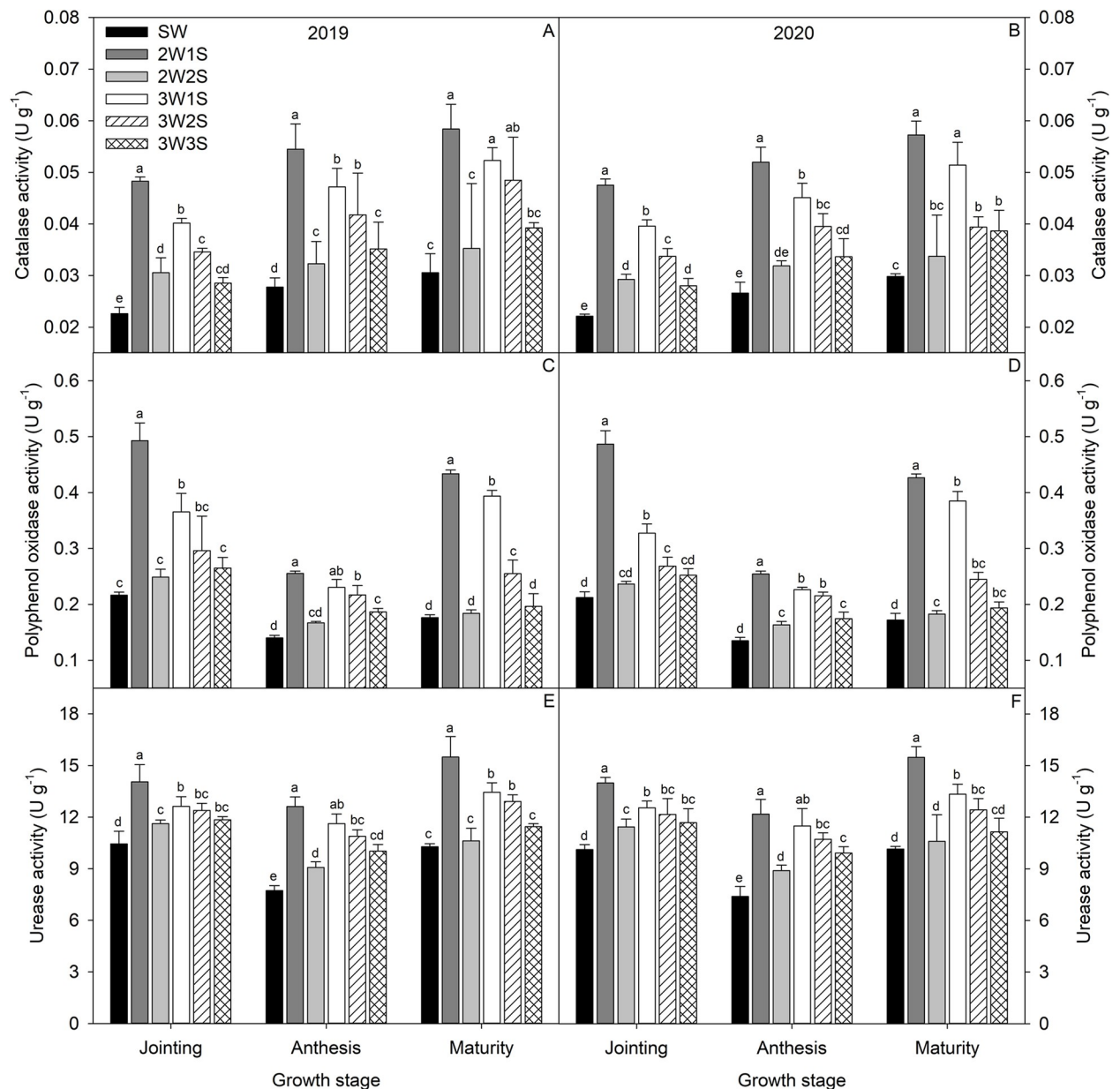


Fig 3. Rhizosphere soil catalase (A, B), polyphenol oxidase (C, D), and urease (E, F) activities of waxy sorghum under different row ratio configurations in waxy sorghum-soybean intercropping systems. Data are expressed as mean $\pm$ SE ($n = 3$). Different letters indicate significant differences among treatments at the $P < 0.05$ level. SW represent sole cropping waxy sorghum. 2W1S, 2W2S, 3W1S, 3W2S, and 3W3S represent two rows of waxy sorghum intercropped with one row of soybean, two rows of waxy sorghum intercropped with two rows of soybean, three rows of waxy sorghum intercropped with one row of soybean, three rows of waxy sorghum intercropped with two rows of soybean, and three rows of waxy sorghum intercropped with three rows of soybean, respectively.

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Discussion

Soil is a complex dynamic environment, which plays an important role in ecosystem function [33, 39]. Soil nutrients are important indicators to evaluate soil quality, which have been widely used to reflect the land productivity and crop productive potential [40, 41]. Intercropping is a sustainable planting pattern, which plays a crucial role in alleviating the land conflict between

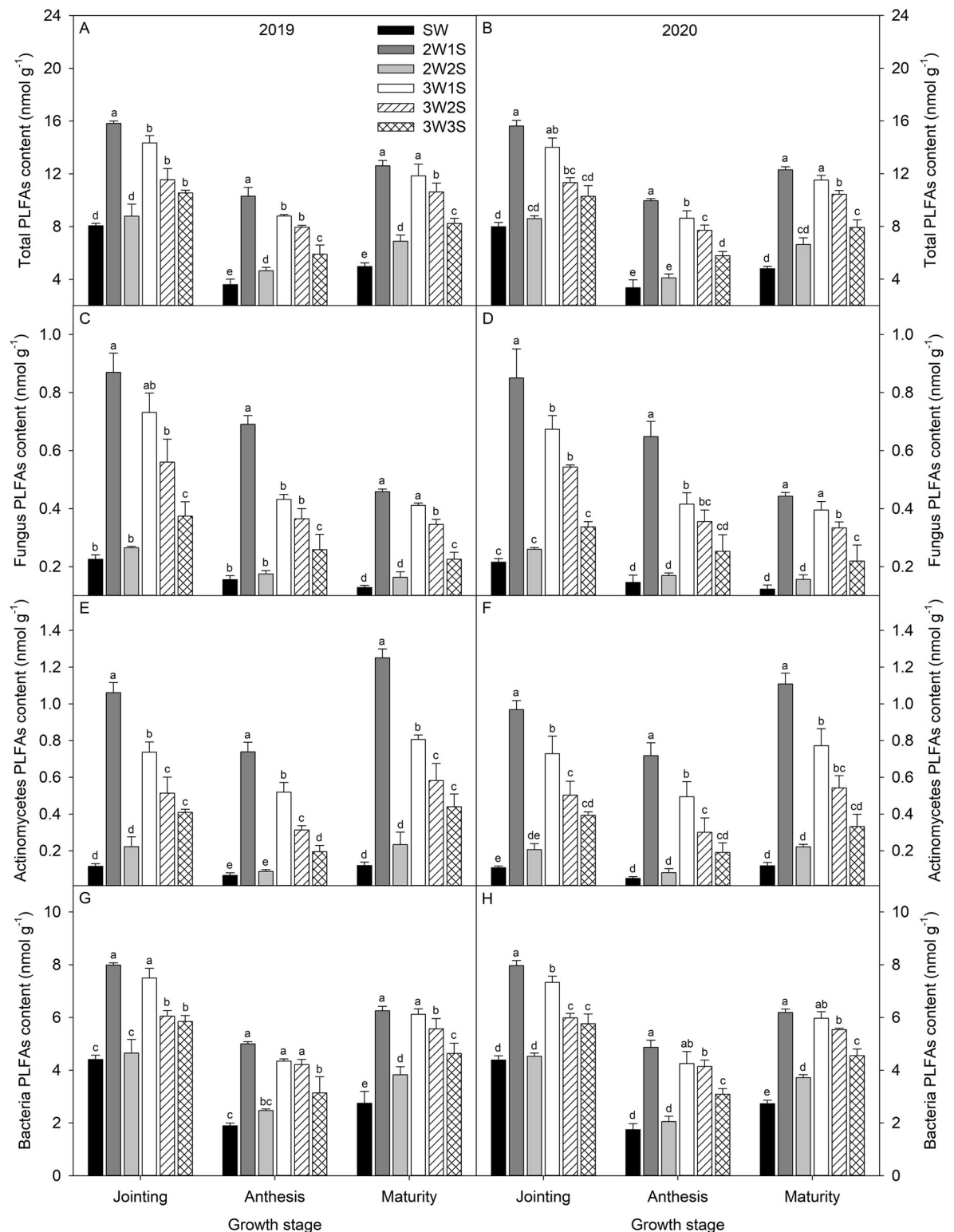


Fig 4. Rhizosphere soil total PLFAs (A, B), fungus PLFAs (C, D), actinomycetes PLFAs (E, F), and bacteria PLFAs (G, H) contents of waxy sorghum under different row ratio configurations in waxy sorghum-soybean intercropping systems. Data are expressed as mean $\pm$ SE ($n = 3$). Different letters indicate significant differences among treatments at the $P < 0.05$ level. SW represent sole cropping waxy sorghum. 2W1S, 2W2S, 3W1S, 3W2S, and 3W3S represent two rows of waxy sorghum intercropped with one row of soybean, two rows of waxy sorghum intercropped with two rows of soybean, three rows of waxy sorghum intercropped with one row of soybean, three rows of waxy sorghum intercropped with two rows of soybean, and three rows of waxy sorghum intercropped with three rows of soybean, respectively.

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Table 1. Rhizosphere soil gram-negative bacteria and gram-positive bacteria PLFAs contents of waxy sorghum under different row ratio configurations in waxy sorghum-soybean intercropping systems.

Year	Treatment	GNB PLFAs content (nmol g ⁻¹)			GPB PLFAs content (nmol g ⁻¹)		
		Jointing stage	Anthesis stage	Maturity stage	Jointing stage	Anthesis stage	Maturity stage
2019	SW	1.76 ± 0.19 d	0.66 ± 0.13 b	0.84 ± 0.24 c	1.56 ± 0.03 e	0.83 ± 0.05 d	1.14 ± 0.08 c
	2W1S	3.09 ± 0.11 a	1.94 ± 0.12 a	2.34 ± 0.21 a	2.82 ± 0.15 a	1.94 ± 0.11 a	2.31 ± 0.12 a
	2W2S	2.01 ± 0.09 cd	0.89 ± 0.16 b	1.14 ± 0.12 c	1.64 ± 0.07 de	1.02 ± 0.07 d	1.52 ± 0.18 bc
	3W1S	2.95 ± 0.12 ab	1.86 ± 0.23 a	2.29 ± 0.18 ab	2.43 ± 0.14 b	1.65 ± 0.12 b	2.22 ± 0.15 a
	3W2S	2.47 ± 0.22 bc	1.72 ± 0.12 a	2.02 ± 0.20 b	1.95 ± 0.07 c	1.34 ± 0.10 c	2.11 ± 0.13 a
	3W3S	2.13 ± 0.16 cd	1.06 ± 0.05 b	1.28 ± 0.31 c	1.79 ± 0.02 cd	1.27 ± 0.05 c	1.65 ± 0.16 b
2020	SW	1.75 ± 0.11 d	0.63 ± 0.17 c	0.76 ± 0.05 c	1.52 ± 0.13 c	0.78 ± 0.08 d	1.06 ± 0.10 c
	2W1S	3.05 ± 0.04 a	1.85 ± 0.19 a	2.31 ± 0.06 a	2.78 ± 0.08 a	1.86 ± 0.12 a	2.24 ± 0.04 a
	2W2S	1.97 ± 0.09 cd	0.84 ± 0.07 c	1.07 ± 0.11 bc	1.62 ± 0.13 c	0.96 ± 0.16 cd	1.46 ± 0.08 b
	3W1S	2.90 ± 0.15 a	1.83 ± 0.30 a	2.25 ± 0.24 a	2.36 ± 0.11 b	1.63 ± 0.17 a	2.13 ± 0.09 a
	3W2S	2.40 ± 0.10 b	1.62 ± 0.12 ab	1.95 ± 0.22 a	1.89 ± 0.16 c	1.29 ± 0.02 b	2.07 ± 0.13 a
	3W3S	2.06 ± 0.06 c	1.03 ± 0.29 bc	1.20 ± 0.16 b	1.73 ± 0.14 c	1.22 ± 0.14 bc	1.63 ± 0.14 b

Data are expressed as mean ± SE (*n* = 3). Different letters indicate significant difference among treatments at the *P* < 0.05 level. SW represent sole cropping waxy sorghum. 2W1S, 2W2S, 3W1S, 3W2S, and 3W3S represent two rows of waxy sorghum intercropped with one row of soybean, two rows of waxy sorghum intercropped with two rows of soybean, three rows of waxy sorghum intercropped with one row of soybean, three rows of waxy sorghum intercropped with two rows of soybean, and three rows of waxy sorghum intercropped with three rows of soybean, respectively.

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crops in Southwest China, especially in Guizhou province [30, 42]. A large number of studies have reported that intercropping improved soil nutrient content, biological activity and stability, increased soil fertilizer supply capacity, improved soil nutrient absorption environment, and ultimately promoted crop growth [15, 22, 27]. Gong et al. [43] reported that proso millet intercropped with mung bean obviously increased the total N, total P, total K, available P, and available K contents of proso millet rhizosphere soil. Similarly, in this study, waxy sorghum intercropped with soybean significantly affected the rhizosphere soil nutrients, and the total N, total P, total K, available N, available P, and available K contents of waxy sorghum rhizosphere soil under all row ratio configurations of waxy sorghum intercropped soybean were higher than those of sole cropping waxy sorghum. Soil organic matter is an important component of

Table 2. Correlation coefficients between soil microbes and soil nutrients as well as soil enzyme activities.

Traits	Total PLFAs content	Fungus PLFAs content	Actinomycetes PLFAs content	Bacteria PLFAs content	Gram-negative bacteria PLFAs content	Gram-positive bacteria PLFAs content
Organic matter content	0.274	0.581 **	0.449 **	0.191	0.293	0.225
Total N content	0.756 **	0.894 **	0.647 **	0.719 **	0.798 **	0.676 **
Total P content	0.308	0.617 **	0.469 **	0.225	0.316	0.272
Total K content	0.926 **	0.853 **	0.931 **	0.896 **	0.873 **	0.928 **
Available N content	0.445 **	0.276	0.765 **	0.404 *	0.319	0.520 **
Available P content	0.449 **	0.673 **	0.592 **	0.367 *	0.478 **	0.401 *
Available K content	0.818 **	0.973 **	0.774 **	0.766 **	0.822 **	0.774 **
Catalase activity	0.594 **	0.609 **	0.864 **	0.527 **	0.493 **	0.643 **
Polyphenol oxidase activity	0.911 **	0.785 **	0.912 **	0.884 **	0.871 **	0.914 **
Urease activity	0.893 **	0.698 **	0.929 **	0.874 **	0.829 **	0.913 **

* and ** indicate significantly correlation at the *P* < 0.05 and *P* < 0.01 level, respectively.

<https://doi.org/10.1371/journal.pone.0288076.t002>

Table 3. Multiple stepwise regression equations between soil microbes and soil nutrients as well as soil enzyme activities.

Multiple stepwise regression equation	Statistic	R^2	F	P
$y_1 = -7.41 + 1.64x_4 - 129.03x_8 + 13.75x_9$	1.55	0.978	232.92	< 0.001
$y_2 = -0.12 - 0.50x_3 + 0.002x_7$	2.34	0.986	582.37	< 0.001
$y_3 = -0.75 + 0.003x_5 + 0.001x_7 + 1.41x_9$	1.85	0.989	477.94	< 0.001
$y_4 = -3.48 + 0.90x_4 - 85.16x_8 + 6.62x_9$	1.48	0.969	161.87	< 0.001
$y_5 = -1.59 + 0.38x_4 - 38.89x_8 + 2.97x_9$	1.42	0.958	120.28	< 0.001
$y_6 = -0.63 + 0.16x_4 + 2.36x_9$	1.33	0.954	167.13	< 0.001

y_1 , total PLFAs content; y_2 , fungus PLFAs content; y_3 , actinomycetes PLFAs content; y_4 , bacteria PLFAs content; y_5 , gram-negative bacteria PLFAs content; y_6 , gram-positive bacteria PLFAs content; x_3 , total P content; x_4 , total K content; x_5 , available N content; x_7 , available K content; x_8 , catalase activity; x_9 , polyphenol oxidase activity.

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soil solid phase, which can improve soil physical properties, promote microbial activities and decomposition of nutrient elements in soil, and increase fertilizer preserving capability and buffer power of soil [32]. The present results showed that the performance of organic matter content among all treatments was 2W1S > 3W1S > 3W2S > 3W3S > 2W2S > SW, indicating that waxy sorghum intercropped with soybean can accelerate the accumulation of organic matter in waxy sorghum rhizosphere soil, which is similar to the previous study about green sword bean intercropped with waxy maize [44]. Thus, waxy sorghum intercropped with soybean can promote the accumulation of organic matter to improve nutritional status in waxy sorghum rhizosphere soil.

Soil enzymes are one of the components of soil, which participate in synthesis and decomposition of humus, hydrolysis and transformation of animals, plants, and microbes residues, and biochemical processes such as redox reactions of organic and inorganic compounds in soil [33, 45]. Soil catalase is a kind of oxidoreductase produced by plants or microbes, which plays an important role in the material circulation of soil ecosystem. It can catalyze the decomposition of hydrogen peroxide in soil into water and oxygen, so as to eliminate and reduce the harm of hydrogen peroxide [46]. Soil polyphenol oxidase is a compound enzyme that is decomposed and released by soil microbes, plant root exudates, and animal and plant residues, which can drive the decomposition and transformation of aromatic compounds in soil, eliminate the adverse effects of aromatic pollutants on soil quality, and thus complete soil environmental remediation [47]. Soil urease is an important part of soil ecosystem and plays an important role in soil material transformation and energy metabolism. As the driving force of soil metabolism, it can catalyze the hydrolysis of urea in soil to ammonia, and is an important index to evaluate the quality of ecological environment and soil fertility [48]. Gong et al. [43] demonstrated that proso millet intercropped with mung bean obviously increased the rhizosphere soil catalase and urease activities of proso millet. Zhou et al. [33] observed that the cucumber rhizosphere soil urease activity under cucumber-onion intercropping system and cucumber rhizosphere soil catalase activity under cucumber-garlic intercropping system were higher than that under sole cropping cucumber. Thus, it was not surprising to observe increases in catalase and urease activities of waxy sorghum rhizosphere soil were found under waxy sorghum-soybean intercropping system in the present study. However, Wu et al. [49] found that continuous sole cropping cucumber increased the polyphenol oxidase activity of cucumber rhizosphere soil, which does not in agreement with in the present study. This inconsistency might be due to the differences in soil ecosystems caused by crops themselves, as waxy sorghum belongs to the Gramineae family and cucumber belongs to the Cucurbitaceae family, which

need to be further studied. These findings imply that waxy sorghum intercropped with soybean can increase enzymes activities and thus improve the quality of waxy sorghum rhizosphere soil.

Soil microbes are an important part of soil ecosystem, and their community composition and quantity changes reflect soil quality and health to a certain extent, and are also the key factors to overcome continuous cropping obstacles [50, 51]. Previous researches showed that rhizosphere soil microbes of crops were significantly affected by intercropping [33, 43, 52]. Therefore, a study on waxy sorghum rhizosphere soil microbes in different row ratio configurations of waxy sorghum intercropped with soybean is necessary to gain a high land use efficiency. PLFAs are present in the membranes of living cells but not in dead cells because they degrade rapidly during cell death, which can accurately estimate the microbial community structure in the rhizosphere soil [53]. In this study, waxy sorghum intercropped with soybean significantly affected the rhizosphere soil total PLFAs content of waxy sorghum and the total PLFAs content of waxy sorghum rhizosphere soil under all row ratio configurations was higher than that of sole cropping waxy sorghum. As far as we know, this is the first time to demonstrate the potential of waxy sorghum-soybean intercropping system to regulate soil microbe populations in Guizhou Karst Mountains ecosystems. The present results showed that all row ratio configurations of waxy sorghum intercropped with soybean possessed higher fungus, actinomycetes, bacteria, gram-negative bacteria, and gram-positive bacteria PLFAs contents of waxy sorghum rhizosphere soil as compared to sole cropping waxy sorghum, which are consistent with previous findings on sugarcane-soybean [54], maize-peanut [55], and corn-soybean [56] intercropping systems. However, Zhou et al. [57] stated that Chinese milk vetch intercropped rape significantly reduced microbes contents in rape rhizosphere soil, including total PLFAs, bacteria, fungus, actinomycetes, gram-negative bacteria, and gram-positive bacteria. This difference might be due to the different accumulation of crop root exudates in the soil caused by the intercropping of crops at different family levels. In the present study, the bacteria PLFAs content under the same growth stage and treatment was the highest among the total PLFAs biomarkers in waxy sorghum rhizosphere soil. One possible reason for this phenomenon might be that the soil bacteria community was more sensitive to the effect of waxy sorghum intercropped with soybean [33]. Another explanation is that waxy sorghum intercropped with soybean promoted the waxy sorghum roots to produce and excrete more organic acids into rhizosphere soil [57]. The present study showed that the gram-negative bacteria and gram-positive bacteria PLFAs contents were roughly the same, suggesting that the nitrogen and carbon cycles of waxy sorghum rhizosphere soil reached an equilibrium state in waxy sorghum intercropped with soybean [56], which requires further studied. This finding demonstrates that waxy sorghum intercropped with soybean can promote the growth and diversity of waxy sorghum rhizosphere soil microbes, and thereby can enhance the soil health status in Guizhou Karst Mountains ecosystems.

Conclusions

Rhizosphere soil properties of waxy sorghum were significantly affected by row ratio configurations of waxy sorghum intercropped soybean. In conclusion, 2W1S treatment was the optimal row ratio configuration of waxy sorghum intercropped with soybean, which can improve the rhizosphere soil quality and promote the sustainable production of waxy sorghum. Future studies will focus on soil microbial diversity based on high-throughput sequencing to better understand the changes in rhizosphere soil microenvironment of waxy sorghum.

Author Contributions

Conceptualization: Mingbo Shao, Can Wang.

Data curation: Mingbo Shao, Can Wang.

Formal analysis: Mingbo Shao, Can Wang, Lingbo Zhou, Fangli Peng, Jie Gao.

Funding acquisition: Mingbo Shao, Can Wang.

Investigation: Can Wang, Fangli Peng, Guobing Zhang, Jie Gao, Siyu Chen, Qiang Zhao.

Methodology: Mingbo Shao, Can Wang.

Project administration: Lingbo Zhou, Guobing Zhang, Qiang Zhao.

Resources: Mingbo Shao.

Supervision: Mingbo Shao.

Validation: Mingbo Shao, Can Wang.

Visualization: Mingbo Shao, Can Wang.

Writing – original draft: Mingbo Shao, Can Wang.

Writing – review & editing: Can Wang, Lingbo Zhou, Fangli Peng.

References

1. Zhou LB, Wang C, GAO X, Ding YQ, Cheng B, Zhang GB, et al. Genome-wide variations analysis of sorghum cultivar Hongyingzi for brewing Moutai liquor. *Hereditas*. 2020; 157:19. <https://doi.org/10.1186/s41065-020-00130-4> PMID: 32410666
2. Wang C, Zhou LB, Zhang GB, Xu Y, Zhang LY, Gao X, et al. Optimal fertilization for high yield and good quality of waxy sorghum (*Sorghum bicolor* L. Moench). *Field Crops Res*. 2017; 203:1–7.
3. Wang C, Zhou LB, Zhang GB, Xu Y, Zhang LY, Gao X, et al. Drought resistance identification and drought resistance indices screening of liquor-making waxy sorghum resources at adult plant stage. *Sci. Agric. Sin*. 2017; 50:1388–1402. Chinese.
4. Wang C, Zhou LB, Gao X, Zhang GB, Cheng B, Cao N, et al. Genetic diversity and population structure analysis of grain-use sorghum based on genotyping by sequencing technology. *J. Plant Genet. Res*. 2019; 20:667–684. Chinese.
5. Wang C, Zhou LB, Zhang GB, Xu Y, Gao X, Jiang N, et al. Effects of sowing time and growing density on agronomic traits, grain yield, and grain quality of waxy sorghum cultivar Hongliangfeng 1. *J. Agric. Sci*. 2019; 11:12–21.
6. Fan FF, Wang JS, Dong EW, Jiao XY, Ding YC, Wu AL, et al. Effect of sorghum continuous cropping on soil microbial community functional diversity. *J. Shanxi Agric. Sci*. 2015; 43:847–853. Chinese.
7. Fan FF, Wang JS, Dong EW, Jiao XY, Ding YC, Wu AL, et al. Effects of sorghum continuous cropping on the growth of sorghum and soil environment. *Soil Fert. Sci. China*. 2016; 3:127–133. Chinese.
8. Yang F, Wang XC, Liao DP, Lu FZ, Gao RC, Liu WG, et al. Yield response to different planting geometries in maize-soybean relay strip intercropping systems. *Agron. J*. 2015; 107:296–304.
9. Nelson WCD, Hoffmann MP, Vadez V, Roetter RP, Whitbread AM. Testing pearl millet and cowpea intercropping systems under high temperatures. *Field Crops Res*. 2018; 217:150–166.
10. Li YJ, Ma LS, Wu PT, Zhao XN, Chen XL, Gao XD. Yield, yield attributes and photosynthetic physiological characteristics of dryland wheat (*Triticum aestivum* L.)/maize (*Zea mays* L.) strip intercropping. *Field Crops Res*. 2020; 248:107656.
11. Tu Y, Yang WY, Liu WG, Yong TW, Jiang LQ, Wang XC. Effects of relay strip intercropping years between flue-cured tobacco and soybean on rhizospheric microbes quantities. *Acta Agron. Sin*. 2015; 41:733–742. Chinese.
12. Bukovsky-Reyes S, Isaac ME, Blesh J. Effects of intercropping and soil properties on root functional traits of cover crops. *Agric. Ecosyst. Environ*. 2019; 285:106614.
13. Feng XM, Yang Y, Ren CZ, Hu YG, Zeng ZH. Effects of legumes intercropping with oat on photosynthesis characteristics of and grain yield. *Acta Agron. Sin*. 2015; 41:1426–1434. Chinese.
14. Ma LK, Dong K, Zhu JH, Dong Y. 2019. Effects of N application on faba bean chocolate spot and canopy microclimate in wheat and faba bean intercropping system. *Chin. J. Appl. Ecol*. 2019; 30:951–960. Chinese.

15. Betencourt E, Duputel M, Colomb B, Desclaux D, Hinsinger P. Intercropping promotes the ability of durum wheat and chickpea to increase rhizosphere phosphorus availability in a low P soil. *Soil Biol. Biochem.* 2012; 46:181–190.
16. Dai J, Qiu W, Wang NQ, Wang TQ, Nakanishi H, Zou YM. From leguminosae/gramineae intercropping systems to see benefits of intercropping on iron nutrition. *Front. Plant Sci.* 2019; 10:605. <https://doi.org/10.3389/fpls.2019.00605> PMID: 31139203
17. Wang XY, Gao YZ. Advances in the mechanism of cereal/legume intercropping promotion of symbiotic nitrogen fixation. *Chin. Sci. Bull.* 2020; 65:142–149. Chinese.
18. Rusinamhodzi L, Corbeels M, Nyamangara J, Giller KE. Maize-grain legume intercropping is an attractive option for ecological intensification that reduces climatic risk for smallholder farmers in central Mozambique. *Field Crops Res.* 2012; 136:12–22.
19. Yang F, Huang S, Gao RC, Liu WG, Yong TW, Wang XC, et al. Growth of soybean seedlings in relay strip intercropping systems in relation to light quantity and red:far-red ratio. *Field Crops Res.* 2014; 155:245–253.
20. Cheng P, Du Q, Liu XM, Zhou L, Hussain S, Lei L, et al. Effects of reduced nitrogen inputs on crop yield and nitrogen use efficiency in a long-term maize-soybean relay strip intercropping system. *PLoS One.* 2017; 12:e0184503. <https://doi.org/10.1371/journal.pone.0184503> PMID: 28910355
21. Sardar H, Khalid Z, Ahsan M, Naz S, Nawaz A, Ahmad R, et al. Enhancement of salinity stress tolerance in lettuce (*Lactuca sativa* L.) via foliar application of nitric oxide. *Plants.* 2023; 12:1115. <https://doi.org/10.3390/plants12051115> PMID: 36903975
22. Ghosh PK, Manna MC, Bandyopadhyay KK, Ajay, Tripathi AK, Wanjarhi RH, et al. Interspecific interaction and nutrient use in soybean/sorghum intercropping system. *Agron. J.* 2006; 98:1097–1108.
23. Egbe OM. Effects of plant density of intercropped soybean with tall sorghum on competitive ability of soybean and economic yield at Otobi, Benue State, Nigeria. *J. Cereals Oilseeds.* 2010; 1:1–10.
24. Baena G, Fera AB, Echevarria C, Monreal JA, García-Mauriño S. Salinity promotes opposite patterns of carbonylation and nitrosylation of C₄ phosphoenolpyruvate carboxylase in sorghum leaves. *Planta.* 2017; 246:1203–1214.
25. Li YS, Yu ZH, Yang SC, Wang GH, Liu XB, Wang CY, et al. Impact of elevated CO₂ on C:N:P ratio among soybean cultivars. *Sci. Total Environ.* 2019; 694:133784.
26. Širić I, Eid EM, Taher MA, El-Morsy MHE, Osman HEM, Kumar P, et al. Combined use of spent mushroom substrate biochar and PGPR improves growth, yield, and biochemical response of cauliflower (*Brassica oleracea* var. botrytis): A preliminary study on greenhouse cultivation. *Horticulturae.* 2022; 8:830.
27. Fan YF, Wang ZL, Liao DP, Raza MA, Wang BB, Zhang JW, et al. Uptake and utilization of nitrogen, phosphorus and potassium as related to yield advantage in maize-soybean intercropping under different row configurations. *Sci. Rep.* 2020; 10:9504. <https://doi.org/10.1038/s41598-020-66459-y> PMID: 32528144
28. Tan Y, Hu FL, Chai Q, Li G, Coulter JA, Zhao C, et al. Expanding row ratio with lowered nitrogen fertilization improves system productivity of maize/pea strip intercropping. *Eur. J. Agron.* 2020; 113:125986.
29. Yin XW, Zhang XC, Wang PH, Cheng H, Wang HM. Optimization of field arrangement of double-cropping glutinous sorghum and soybean intercropping pattern. *Agric. Sci. Technol.* 2015; 16:1110–1114.
30. Wang C, Zhou LB, Zhang GB, Gao J, Peng FL, Zhang CL, et al. Responses of photosynthetic characteristics and dry matter formation in waxy sorghum to row ratio configurations in waxy sorghum-soybean intercropping systems. *Field Crops Res.* 2021; 263:108077.
31. Wang C, Zhou LB, Gao J, Zhang GB, Peng FL, Zhang CL, et al. Changes in nutrient accumulation and transportation of waxy sorghum in waxy sorghum-soybean intercropping systems under different row ratio configurations. *Front. Plant Sci.* 2022; 13:921860. <https://doi.org/10.3389/fpls.2022.921860> PMID: 35937366
32. Yang LS, Zhang YT, Huang XC, Zhang YQ, Zhao YN, Shi XJ. Effects of long-term application of chloride containing fertilizers on the biological fertility of purple soil under a rice-wheat rotation system. *Sci. Agric. Sin.* 2016; 49:686–694. Chinese.
33. Zhou XG, Yu GB, Wu FZ. Effects of intercropping cucumber with onion or garlic on soil enzyme activities, microbial communities and cucumber yield. *Eur. J. Soil Biol.* 2011; 47:279–287.
34. Wang QY, Zhou DM, Cang L. Microbial and enzyme properties of apple orchard soil as affected by long-term application of copper fungicide. *Soil Biol. Biochem.* 2009; 41:1504–1509.
35. Luo XS, Fu XQ, Yang Y, Cao P, Peng SB, Chen WL, et al. Microbial communities play important roles in modulating paddy soil fertility. *Sci. Rep.* 2016; 6:20326. <https://doi.org/10.1038/srep20326> PMID: 26841839

36. Zhang ZJ, Qu YY, Li SZ, Feng K, Wang S, Cai WW, et al. Soil bacterial quantification approaches coupling with relative abundances reflecting the changes of taxa. *Sci. Rep.* 2017; 7:4834.
37. Chen K, Xu XN, Peng J, Feng XJ, Li YP, Zhan XM, et al. Effects of biochar and biochar-based fertilizer on soil microbial community structure. *Sci. Agric. Sin.* 2018; 51:1920–1930. Chinese.
38. Wan XH, Huang ZQ, He ZM, Yu ZP, Wang MH, Davis MR, et al. Soil C: N ratio is the major determinant of soil microbial community structure in subtropical coniferous and broadleaf forest plantations. *Plant Soil.* 2015; 387:103–116.
39. Acosta-Martínez V, Burow G, Zobeck TM, Allen VG. Soil microbial communities and function in alternative systems to continuous cotton. *Soil Sci. Soc. Am. J.* 2010; 74:1181–1192.
40. Wang ZG, Bao XG, Li XF, Jin X, Zhao JH, Sun JH, et al. Intercropping maintains soil fertility in terms of chemical properties and enzyme activities on a timescale of one decade. *Plant Soil.* 2015; 391:265–282.
41. Deng L, Yan WM, Zhang YW, Shuangguan ZP. Severe depletion of soil moisture following land-use changes for ecological restoration: Evidence from northern China. *Forest Ecol. Manag.* 2016; 366:1–10.
42. Huang CJ, Zhao SY, Wang LC, Wang JC, Zhao Y, Cai YM, et al. Effect of potato/maize intercropping on photosynthetic characteristics and yield in two potato varieties. *Acta Agron. Sin.* 2013; 39:330–342. Chinese.
43. Gong XW, Liu CJ, Li J, Luo Y, Yang QH, Zhang WL, et al. Responses of rhizosphere soil properties, enzyme activities and microbial diversity to intercropping patterns on the Loess Plateau of China. *Soil Till. Res.* 2019; 195:104355.
44. Zhang JY, Ning TY, Feng YP, Zhang XP, Sun T, Li ZJ. Effects of intercropping with waxy maize on crop yield and soil properties of the continuous vegetable field. *Sci. Agric. Sin.* 2013; 46:1994–2003. Chinese.
45. Saha S, Prakash V, Kundu S, Kumar N, Mina BL. Soil enzymatic activity as affected by long term application of farm yard manure and mineral fertilizer under a rainfed soybean-wheat system in N-W Himalaya. *Eur. J. Soil Biol.* 2018; 44:309–315.
46. Zhang WJ, Yang LL, Wang JH, Zhu LS, Wang J, Mao SS. Effect of combined pollution of three antibiotics and Cu on soil catalase activity. *J. Agric. Resour. Environ.* 2020; 37:135–143. Chinese.
47. Ma ZL, Zhao WQ, Liu M. Responses of polyphenol oxidase and catalase activities of rhizosphere and bulk soils to warming during the growing season in an alpine scrub ecosystem. *Chin. J. Appl. Ecol.* 2019; 30:3681–3688. Chinese.
48. Song T, Yin JH, Hu ZP, Wang LL, Zhang Q, Chen Q, et al. Characteristics of urease/nitrification inhibitors in reducing nitrogen losses in farmland soils. *J. Agric. Resour. Environ.* 2021; 38:585–597. Chinese.
49. Wu FZ, Meng LJ, Wang XZ. Soil enzyme activities in vegetable rotation and continuous cropping system of under shed protection. *Plant Nutr. Fert. Sci.* 2006; 12:554–558. Chinese.
50. Chaudhary P, Xu M, Ahamad L, Chaudhary A, Kumar G, Adeleke BS, et al. Application of synthetic consortia for improvement of soil fertility, pollution remediation, and agricultural productivity: A review. *Agron.* 2023; 13:643.
51. Chen DM, Duan YQ, Yang YH, Jin Y, Huang JG, Yuan L. Effects of long-term fertilization on flue-cured tobacco soil nutrients and microorganisms community structure. *Sci. Agric. Sin.* 2014; 47:3424–3433. Chinese.
52. Dai CC, Chen Y, Wang XX, Li PD. Effects of intercropping of peanut with the medicinal plant *Atractylodes lancea* on soil microecology and peanut yield in subtropical China. *Agroforest. Syst.* 2013; 87:417–426.
53. Kong AYY, Scow KM, Córdova-Kreylos AL, Holmes WE, Six J. Microbial community composition and carbon cycling within soil microenvironments of conventional, low-input, and organic cropping systems. *Soil Biol. Biochem.* 2011; 43:20–30. <https://doi.org/10.1016/j.soilbio.2010.09.005> PMID: 22267876
54. Li XP, Mu YH, Cheng YB, Liu XG, Nian H. Effects of intercropping sugarcane and soybean on growth, rhizosphere soil microbes, nitrogen and phosphorus availability. *Acta Physiol. Plant.* 2013; 35:1113–1119.
55. Li QS, Wu LK, Chen J, Muhammad AK, Luo XM, Lin WX. Biochemical and microbial properties of rhizospheres under maize/peanut intercropping. *J. Integr. Agric.* 2016; 15:101–110.
56. Zaeem M, Nadeem M, Pham TH, Ashiq W, Ali W, Gilani SSM, et al. The potential of corn-soybean intercropping to improve the soil health status and biomass production in cool climate boreal ecosystems. *Sci. Rep.* 2019; 9:13148. <https://doi.org/10.1038/s41598-019-49558-3> PMID: 31511594
57. Zhou Q, Chen J, Xing Y, Xie XY, Wang LC. Influence of intercropping Chinese milk vetch on the soil microbial community in rhizosphere of rape. *Plant Soil.* 2019; 440:85–96.